



STUDENT PROJECTS

PROJECT PLAN

Department: EC Semester: 6th Academic Year: 2026– 27

A. Proposed Title of the Project:

Universal Asynchronous Receiver and Transmitter (UART)

B. Name(s) of guide(s): Prof. / Dr. / Mr. / Ms.

C. Name of Team Members (Strictly not more than four students in a batch):

USN	Name	
4VP23EC050	Mr. THIYANAND M	Team Leader
4VP23EC005	Ms. CHAITHANYA	Team Member
4VP23EC029	Ms. PREETHA S M	Team Member
4VP23EC010	Ms. FATHIMATH ANEESHA	Team Member

D. Date of commencement of project:

E. Probable date of completion of the project:

F. If this project is an improvement / extension of an earlier project, please give the title of earlier project and the year in which it was carried out. Specify the improvements / modifications clearly:

Not Applicable

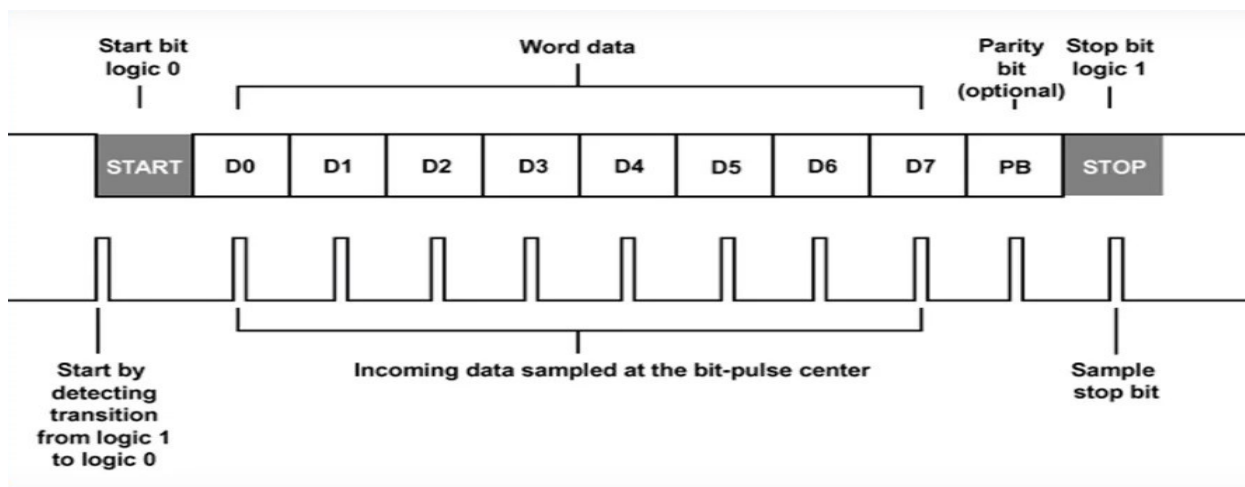
G. Objectives of the project highlighting its importance (not more than half a page):

- To design and implement a Universal Asynchronous Receiver and Transmitter (UART) using basic digital electronic components on a breadboard.
- To understand the concept of asynchronous serial communication without using a common clock between transmitter and receiver.
- To study the working of UART transmitter and receiver including start bit, data bits, and stop bit.
- To analyse baud rate generation and synchronization in serial communication.
- To bridge theoretical communication concepts with practical hardware implementation using UART



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H. Block diagram and Methodology (Not more than a page):



The device operates in the following sequence:

1. Power Supply Setup
A regulated 5V power supply is designed and tested to ensure stable operation of all ICs used in the UART circuit.
2. Clock / Baud Rate Generation
A clock or baud rate generator circuit is implemented to control the speed of serial data transmission and reception.
3. UART Transmitter Design
The transmitter circuit converts parallel data into serial form by adding start and stop bits using shift registers and control logic.
4. UART Receiver Design
The receiver circuit detects the start bit, samples incoming serial data at the selected baud rate, and converts it back into parallel data.
5. Data Registers and Buffers
Registers are used to temporarily store transmitted and received data during communication.
6. Control Logic Implementation
Control logic is designed to manage data loading, shifting, and synchronization between the transmitter and receiver.
7. Simulation and Verification
The UART design can be simulated and verified using tools such as Proteus or ModelSim before or alongside breadboard implementation.
8. Testing and Output Verification
The complete UART system is tested on a breadboard and verified for correct data transmission and reception using LEDs or interfacing devices.

I. Expected Outcome of the project:

Prepared by:

Checked by:

HOD

Nehru Nagar, Puttur - 574 203, DK, Karnataka State – INDIA.

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- Successful design and implementation of a functional UART system.
- Correct transmission and reception of serial data using asynchronous communication.
- Proper generation and detection of start, data, and stop bits.
- Reliable operation of UART at the selected baud rate.
- Improved practical understanding of serial communication used in embedded systems.

J. Application of the project:

- Used for serial communication in microcontroller-based systems.
- Widely applied in embedded and IoT systems for data exchange.
- Used in communication between computers and peripheral devices.
- Serves as an educational model to understand UART protocol and serial communication.
- Useful for digital electronics and communication laboratory experiments.

K. Does the project proposed is relevant to any of the Industry or Institution in and around your area (Yes / No):

No

L. If yes, please provide details of the industry / institution and contact details:

M. Budget:

Materials Cost: 5500/-

Labour Charges:

Any other cost:

Source of Funds:

Self / Industry / Others

N. Expected Completion Time:

Expected Time for Major Activities

Schedule

Synopsis (Project Plan) submission to the Department

Presentation of synopsis

Last date for re-submission (if applicable) & review

Submission of Reports

Internal Viva and/or Project Exhibition

O. Authors:

Student(s) / Author(s)

Signature with date

1. THIYANAND M

2. CHAITHANYA

3. PREETHA S M

4. FATHIMATH ANEESHA

Checked By:

Signature with date:

P. Guidance:

Guide (s) allotted:

Signature (s) with date

1. Guide: Prof. / Dr. / Mr. / Ms.

2. Co-Guide (if any): Prof. / Dr. / Mr. / Ms.

Prepared by:

Checked by:

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